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ADA TECH SURVEY

Geotechnical Assessment & Recommendations for Secure Perimeter Development

Plot Size: 100 ft × 143 ft

Location: Adjacent to Ada Technical School, Greater Accra Region, Ghana

Prepared for: Mr Eric Teye Abbey

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Date: 8th December, 2025

Following our recent site visit and preliminary soil observations, we are pleased to submit this professional report and actionable recommendations for the immediate securing and future development of your plot near Ada Technical School (herein referred to as the “Ada Tech Survey” site).

1. Site Conditions & Soil Characteristics

The land is a typical coastal-plain wetland environment dominated by expansive clay soil at the surface. This clay is highly plastic: it becomes extremely soft and unstable when saturated with water and hardens and cracks when dry. Such behaviour creates serious challenges for direct construction without proper ground improvement and dewatering measures.

Visual evidence from site photographs confirms:

- Surface ponding and erosion gullies after light rain
- High grass cover with patchy bare clay areas
- Proximity to a seasonal high water table



2. Immediate Recommended Actions

A. Trial Holes (Test Pits)

To obtain an accurate soil profile and confirm groundwater levels, we will excavate three (3) trial holes at the following locations:

- One at the north-east corner
- One at the south-west corner
- One at the centre of the plot

Each hole will be dug to a depth of 5–7 feet (or until a more stable stratum is encountered). Soil samples will be taken at 1–2 ft intervals and assessed. This depth has been chosen because the active zone of expansive clay in this area typically extends 5–6 ft below surface; going beyond this ensures we capture the full behaviour of the soil that will affect your foundations.

B. Perimeter Security Wall (Short-Term Priority)

To protect and demarcate the land immediately, we propose constructing a 5 ft high perimeter block wall with the following specifications that have been specially designed for this difficult clay soil:

- Foundation trench: 2 ft 6 in deep × 18 in wide for the wall
- Column footings: 3 ft deep × 2 ft × 2 ft pad
- Columns: 6 in × 6 in reinforced concrete at 10 ft centres
→ 4Y12 main bars + Y8 stirrups @ 7 in centres
- First five courses: 6 in sandcrete blocks laid flat (stronger base) using 42.5R cement + sharp river sand mortar
- Damp-proof course (DPC) membrane after the fifth course
- Reinforced concrete ring beam cast on top of the fifth course to tie all columns together
- Concrete grade for all works: C25 (25 N/mm²)

This system will resist settlement and lateral movement even when the clay becomes saturated.

C. Permanent Dewatering System (Cesspool + Perimeter Drains)

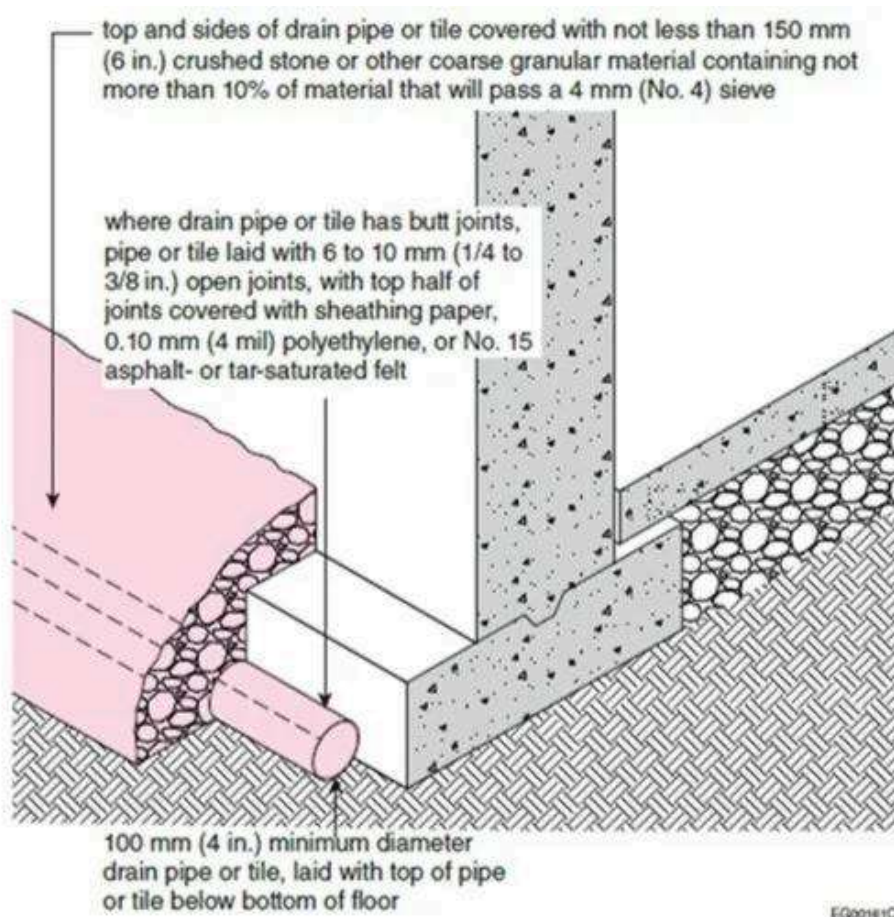
Because this is a wetland plot, a properly designed drainage system is non-negotiable. We recommend the following proven system (illustrated in the attached diagrams):

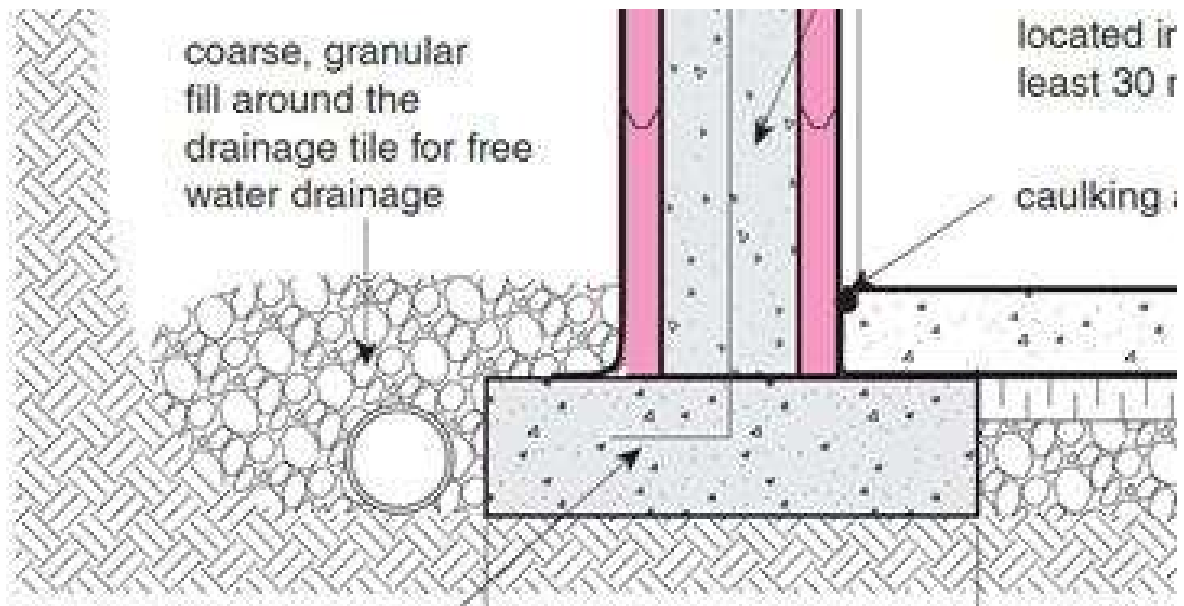
1. Excavate a narrow trench along the inside of the perimeter wall footing (after the wall is built).
2. Lay 4–6 in perforated PVC pipes with holes facing downward.
3. Surround the pipe completely with clean 20–40 mm granite chippings (minimum 150 mm cover on all sides).
4. Wrap the entire gravel envelope with geotextile fabric to prevent clay particles from clogging the system.
5. Slope the pipes gently (minimum 1:100) towards a central concrete-lined cesspool (10–15 ft deep × 6–8 ft diameter).
6. Install a submersible pump in the cesspool for periodic or automatic water removal.

This system will keep the ground beneath your future building permanently dry and will continue to function even after roads and gutters are constructed around the site.

Attached Diagrams (for your easy understanding):

1. Standard detail of perforated pipe in gravel trench with geotextile wrap
2. Cross-section showing drain placement adjacent to footing
3. Coarse granular backfill detail for free water flow





3. Next Steps & Timeline (Weather Permitting)

1. Day 1–3 → Excavate and log the three trial holes
2. Week 2 → Laboratory testing of samples
3. Week 3–6 → Construct perimeter security wall with integrated drainage system
4. Week 7 → Final dewatering system commissioning and handover of secured plot

We remain confident that with these measures, your land will be fully secured, protected from encroachment, and perfectly prepared for any future building works — whether residential, educational extension, or light commercial.

Should you wish to proceed, kindly give us your written approval and we will mobilise to site within 48 hours.

Thank you for trusting Modular Automation & Design with this important project. We look forward to delivering a safe, durable, and future-proof solution for you.